SWINBURNE VIETNAM

HO CHI MINH CAMPUS

COS30018 - COS30018-Intelligent Systems

Oral Defense week 11

INSTRUCTOR: MR. KY TRUNG PHAM

STUDENT NAME: LÊ TIẾN DŨNG-SWS00683

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Contents

Oral Defense week 11 3

1. Based on your findings, do you believe AI can truly "predict" the stock market, or is it merely following historical trends? Justify your answer with your results. 3

2. How does a Transformer's self-attention differ from an LSTM, and why might it be better for long-term dependencies? 3

3.(Lecture 10 Concept): In Lecture 10, we discussed Knowledge Representation and Ontologies. How could you use an ontology of the financial market (e.g., explicitly mapping that "Apple" and "Microsoft" are both in the "Tech Sector") to improve your multivariate prediction model? 4

Oral Defense week 11

1. Based on your findings, do you believe AI can truly "predict" the stock market, or is it merely following historical trends? Justify your answer with your results.

My answer is: AI can learn useful patterns from historical data, but it cannot truly predict the stock market with certainty.

In my project, the models clearly learned something useful, because the ensemble improved performance compared with weaker components. In C.6, the best ensemble used GRU + SARIMA with $w_{dl} = 0.8$and achieved ENS_MAE ≈ 2.7925**,** while SARIMA alone had much higher error at about MAE ≈ 8.885. That shows the model extracted some predictive signal from past data.

But at the same time, the results also show limitation: the system is mainly learning patterns in historical price behavior, not understanding future market shocks, news surprises, policy changes, or human behavior in a complete way. In C.7, even the best directional classifier only reached about F1 ≈ 0.62, which is useful but far from perfect. That means AI can provide a probabilistic edge, but not a guaranteed forecast.

So, the balanced answer is:

AI in my project is not “seeing the future.” It is learning repeatable structures from historical prices and, in C.7, some sentiment information. That can improve forecasting, but it is still mostly pattern-following under uncertainty, not perfect market prediction.

2. How does a Transformer’s self-attention differ from an LSTM, and why might it be better for long-term dependencies?

The LSTM is still a recurrent model. It processes the sequence step by step and carries memory forward through hidden state and cell state. The Week 11 lecture explains that LSTM uses gates to control what information to retain or discard over time.

A Transformer works differently. Instead of processing only one step after another through recurrence, it uses self-attention to decide which parts of the input should be focused on. In the lecture slides, the attention diagram literally shows the model asking: “What part of the input should we focus?” and assigning different attention weights to different words. It also adds positional encoding so the model still knows the order of the sequence.

Why this can be better for long-term dependencies:

- RNNs and even LSTMs still process sequences sequentially, so long histories can be harder and slower to learn. The lecture explicitly lists RNN limitations such as slow computation due to no parallelization and no long memory, and even notes LSTMs are slow to train.
- With self-attention, a Transformer can directly connect one time step to another far-away time step in the same sequence, instead of passing information through many recurrent steps.
- It can also process positions more in parallel than an LSTM.

So for stock data, a Transformer might be better if long-range dependencies matter, for example when price behavior depends on events or patterns that happened much earlier. I would still say “might,” because stock data is noisy and Transformers usually need more data and tuning than LSTMs.

In shorter word:

LSTM remembers information through recurrent hidden states and gates, while a Transformer uses self-attention to directly weight the relevance of all positions in the sequence. Because of that, Transformers can model long-range relationships more directly and with better parallelization, which may help with long-term dependencies in market data.

3.(Lecture 10 Concept): In Lecture 10, we discussed Knowledge Representation and Ontologies. How could you use an ontology of the financial market (e.g., explicitly mapping that "Apple" and "Microsoft" are both in the "Tech Sector") to improve your multivariate prediction model?

In my current multivariate model, I already use multiple market features such as Open, High, Low, Close, Adj Close, and Volume as structured numeric inputs.

An ontology could improve this by adding explicit financial relationships, not just raw numbers. For example, an ontology could represent facts like:

- Apple and Microsoft are both in the Tech Sector
- CBA belongs to the Banking / Financial Sector
- a company belongs to a country, exchange, industry, and supply-chain group
- macro variables such as interest rates affect banks differently from mining or tech stocks

That would help because the model would no longer treat all variables as isolated columns. Instead, it would know which assets are related, comparable, or likely to move together.

In practice, I could use an ontology in three ways:

1. Feature grouping
   Add sector-level or industry-level features, such as sector average return, sector volatility, or sector sentiment.

2. Relational features
   If one bank stock moves strongly, that information could be propagated to related banking stocks because the ontology says they belong to the same sector.

3. Smarter multivariate selection
   Instead of choosing extra variables manually, I could select related assets based on ontology links like sector, geography, or business type.

So, the main value of ontology is that it injects domain knowledge into the model. It helps the model understand that some stocks are related by meaning, not only by past price correlation.

In shorter word:

My current multivariate model uses numerical market features, but an ontology would add structured financial knowledge, such as sector and industry relationships. That could help me create better features, choose more relevant related variables, and improve prediction by telling the model which companies are meaningfully connected.